WORKED SOLUTION

STEP I 2012, Question 2

Pure • Curve Sketching and Roots

STEP I 2012, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q2

The key idea

Counting the real roots of $p(x) = 0$ is the same as counting where the graph of $y = p(x)$ meets the x-axis, or, after moving the constant across, where a fixed curve meets a horizontal line. So the whole question is about knowing the shape of a quartic: where it turns, and how high or low it is when it does. In part (i) the quartic is a perfect square; in part (ii) the condition on the derivatives makes $\frac{dy}{dx}$ factorise with a repeated root, which leaves only one turning point.

Solution

Part (i)

Notice first that

$$y = x^4 - 6x^2 + 9 = (x^2 - 3)^2,$$

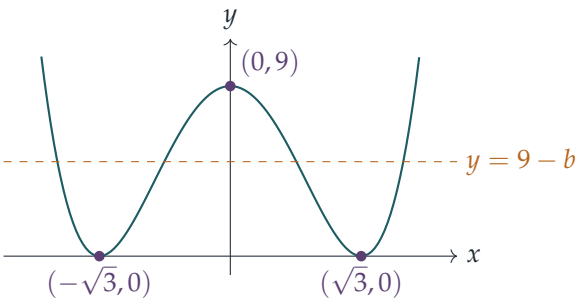
so $y \geq 0$ everywhere, with $y = 0$ exactly at $x = \pm\sqrt{3}$. The curve is even (symmetric in the y-axis), and it crosses the y-axis at $(0, 9)$. Differentiating,

$$\frac{dy}{dx} = 4x^3 - 12x = 4x(x^2 - 3),$$

which vanishes at $x = 0$ and $x = \pm\sqrt{3}$. The stationary points are therefore

$$(0,\, 9) \text{ (a local maximum)}, \qquad (\sqrt{3},\, 0) \text{ and } (-\sqrt{3},\, 0) \text{ (minima)}.$$

The minima are clear because $y \geq 0$ and $y = 0$ there; and $\frac{d^2y}{dx^2} = 12x^2 - 12$ is $-12 < 0$ at $x = 0$, so that point is a maximum. As $x \to \pm\infty$, $y \to +\infty$. The curve is a "W" that touches the x-axis at $x = \pm\sqrt{3}$.



For the root count, add 9 to both sides:

$$x^4 - 6x^2 + b = 0 \iff (x^2 - 3)^2 = 9 - b.$$

So n is the number of times the horizontal line $y = 9 - b$ meets the curve we have just sketched. Reading off the picture as the line moves from high to low:

- $9 - b > 9$, i.e. $b < 0$: the line is above the maximum and meets only the two outer arms, so $n = 2$.
- $9 - b = 9$, i.e. $b = 0$: the line passes through the maximum, giving $x = 0$ and two outer points; indeed $x^2(x^2 - 6) = 0$ gives $x = 0, \pm\sqrt{6}$. So $n = 3$.
- $0 < 9 - b < 9$, i.e. $0 < b < 9$: the line cuts each of the four monotonic pieces once, so $n = 4$.
- $9 - b = 0$, i.e. $b = 9$: the line touches at the two minima only, $x = \pm\sqrt{3}$, so $n = 2$.
- $9 - b < 0$, i.e. $b > 9$: the line lies below the curve, so $n = 0$.

Collecting these:

$$\text{(a) } n = 0 \iff b > 9; \quad \text{(b) } n = 1 \text{ for no } b; \quad \text{(c) } n = 2 \iff b < 0 \text{ or } b = 9;$$

$$\text{(d) } n = 3 \iff b = 0; \quad \text{(e) } n = 4 \iff 0 < b < 9.$$

That $n = 1$ is impossible also follows from symmetry: if x is a root then so is $-x$, so a single root would have to be $x = 0$, which forces $b = 0$, and then there are three roots.

Part (ii)

Now $y = x^4 - 6x^2 + ax + b$, so

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4x^3 - 12x + a, \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 12x^2 - 12.$$

The second derivative vanishes only at $x = \pm 1$. At $x = 1$ the first derivative is $a - 8$, and at $x = -1$ it is $a + 8$. So both vanish at the same point exactly when

$$a = 8 \text{ (at } x = 1) \qquad \text{or} \qquad a = -8 \text{ (at } x = -1).$$

Case $a = 8$**.** Here

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 4(x^3 - 3x + 2) = 4(x-1)^2(x+2),$$

where we found the factor $(x-1)^2$ because we built $x = 1$ to be a root of both y' and y'', i.e. a repeated root of y'.

? **Ask yourself** How many turning points does this curve actually have?

Since $(x-1)^2 \geq 0$, the sign of $\frac{\mathrm{d}y}{\mathrm{d}x}$ is the sign of $x+2$. So y is strictly decreasing for $x < -2$ and strictly increasing for $x > -2$ (the gradient is zero at $x = 1$ but does not change sign: that is a stationary point of inflection, not a turning point). There is just one turning point, a global minimum at $x = -2$, where

$$y(-2) = 16 - 24 - 16 + b = b - 24.$$

A curve that falls to a single minimum and then rises meets the x-axis twice if the minimum is below the axis, once (touching) if it is on the axis, and not at all if it is above. Hence the number of distinct real roots is

$$\begin{cases} 2 & \text{if } b < 24, \\ 1 & \text{if } b = 24, \\ 0 & \text{if } b > 24. \end{cases}$$

(For $b = 24$ we can see the double root directly: $x^4 - 6x^2 + 8x + 24 = (x+2)^2(x^2 - 4x + 6)$, and the quadratic has discriminant $16 - 24 < 0$.)

Case $a = -8$**.** Replacing x by $-x$ turns $x^4 - 6x^2 - 8x + b$ into $x^4 - 6x^2 + 8x + b$, so the curve is the reflection of the previous one in the y-axis. The single minimum is now at $x = 2$, with value $b - 24$, and the number of roots is the same: 2 if $b < 24$, 1 if $b = 24$, 0 if $b > 24$.

Part (iii)

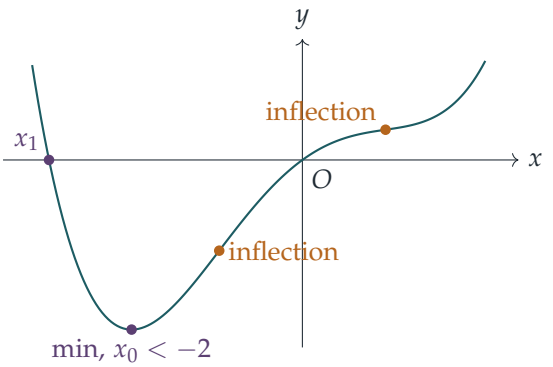
Take $y = x^4 - 6x^2 + ax$ with $a > 8$.

Stationary points. $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ means $x^3 - 3x = -\frac{a}{4}$. The cubic $g(x) = x^3 - 3x$ has a local maximum $g(-1) = 2$ and a local minimum $g(1) = -2$, and $g(-2) = -2$. Since $-\frac{a}{4} < -2$, the horizontal line at height $-\frac{a}{4}$ lies below the local minimum of g, so it meets g exactly once, on the left-hand branch where $x < -2$. So there is exactly one stationary point, at some $x_0 < -2$. The gradient is negative to its left and positive to its right, so it is the minimum.

Crossings of the x-axis. $y = x(x^3 - 6x + a)$, so $x = 0$ is one crossing, and the others are roots of $h(x) = x^3 - 6x + a$. This cubic has its local minimum at $x = \sqrt{2}$, where $h(\sqrt{2}) = a - 4\sqrt{2} > 0$ because $a > 8 > 4\sqrt{2}$. So h has only one real root, and it is negative because $h(0) = a > 0$. In fact $h(-2) = a + 4 > 0$, and h is increasing for $x < -\sqrt{2}$, so this root x_1 satisfies $x_1 < -2$. The curve crosses the axis at x_1 and at the origin, and is negative between them; the minimum lies in this interval.

Other features. At the origin the gradient is $a > 0$. The second derivative $12(x^2 - 1)$ changes sign at $x = \pm 1$, so there are points of inflection at $(-1, -a - 5)$ and $(1, a - 5)$; the gradient there is $a + 8$ and $a - 8$, both positive, so the curve is still rising through them. It is concave down for $-1 < x < 1$ and concave up outside. As $x \to \pm\infty$, $y \to +\infty$.

The sketch shows the case $a = 10$, where $x_0 \approx -2.05$, the minimum value is about -28, and $x_1 \approx -3.05$.



Checks

- In part (i), $b = 5$ should give four roots: $x^2 = 3 \pm 2$, so $x = \pm 1, \pm\sqrt{5}$. It does.
- In part (ii), $b = 24$ gives the factorisation $(x + 2)^2(x^2 - 4x + 6)$, confirming the single (double) root at the minimum.
- Numerically, we found all complex roots of the quartics for many values of b either side of 0, 9 and 24, and counted the distinct real ones; every count matched. For random a between 8 and 200, the curve in (iii) always had one stationary point and two axis crossings, with $x_1 < x_0 < -2$. For $a = 10$, $x_0 \approx -2.054$ and $y(x_0) \approx -28.05$.

Common slips

- Counting the double roots at $x = \pm\sqrt{3}$ (when $b = 9$) as four roots: the question asks for *distinct* values of x.
- Treating the point where $y' = y'' = 0$ in part (ii) as a turning point. Because y' has a squared factor there, its sign does not change, so it is a stationary point of inflection.
- Forgetting $a = -8$: $y'' = 0$ has two solutions, and each gives a value of a.

Beyond STEP

The way the number of real roots jumps as b passes through 0, 9 or 24 is a small example of a *bifurcation*: the answer changes only when a horizontal line passes through a stationary value. The special case $y' = y'' = 0$ in part (ii) is where two stationary points merge and vanish, which is the simplest "catastrophe" (the fold) and the reason the quartic in (iii) has only one turning point once $a > 8$. The same reasoning, made rigorous by the intermediate value theorem and monotonicity, is the start of real analysis; Abbott's *Understanding Analysis* is a good place to see it done carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.